NNN-D-340a
December 26, 1962

SUPERSEDING
Int. Fed. Spec. NNN-D-00340 (ARMY-CmlC)
June 30, 1961, and
Fed. Spec. DD-D-420
July 18, 1955

FEDERAL SPECIFICATION

DISH, EVAPORATING, NONMETALLIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers evaporating dishes suitable for laboratory use. Platinum evaporating dishes are excluded.

1.2 **Classification.**

1.2.1 *Types and nominal capacities.* Evaporating dishes shall be of the following types and nominal capacities, as specified (see 6.2):

Type I—Porcelain.

Capacity, milliliters

35
70
90
120
150
250
525
1285
2100

Type II—Fuzed silica, opaque or transparent.

Capacity, milliliters.

200

Type III—Borosilicate glass.

Capacity, milliliters.

90
195

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

DD-G-541—Glass (Laboratory).

UU-P-268—Paper, Kraft, Untreated, Wrapping.

UU-T-111—Tape; Paper, Gummed (Sealing and Securing).

PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

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PPP-C-843—Cushioning Material, Cellulosic.

PPP-T-60—Tape; Pressure Sensitive Adhesive, Waterproof—For Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 626—Glassware; Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-10547—Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Classification Committee Publication:

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

American Society For Testing And Materials Standard (ASTM):

C 146-43—Chemical Analysis of Glass Sand.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Porcelain.* Type I evaporating dishes shall be a high quality vitreous ceramic whiteware having an average thermal coefficient of linear expansion not greater than 3.6×10^{-6} centimeter (cm)/cm/1°C. from 20° to 200° C. and gradually increasing to not greater than 4.7×10^{-6} cm/cm/1°C. at 1000°C.

3.1.1.1 *Thermal shock resistance.* No cracking shall occur when evaporating dishes are tested as specified in 4.6.1.1.

3.1.1.2 *Glaze.* The glaze shall offer a high degree of inertness and have a thermal coefficient of expansion approximately equal to that of the porcelain body. All dishes from 35-to 250-ml capacity, inclusive, shall be glazed inside and outside with the exception of

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the rim. Dishes of 525- to 2100-ml capacity, inclusive, shall be glazed inside and only for a short distance below the rim on the outside.

3.1.1.2.1 *Glaze adhesion.* Permanent adhesion of the glaze to another glazed porcelain surface of the same material shall not occur below 1250°C. when tested as specified in 4.6.2.

3.1.1.2.1.1 *Permanent adhesion.* Adhesion shall be considered permanent when, after cooling to room temperature, the touching surfaces are annealed as one or, the touching surfaces separate with a portion of one surface adhering to and becoming part of the other surface.

3.1.1.2.2 *Resistance to chemical attack.* Resistance to chemical attack shall be measured by loss of weight by boiling in a solution of sodium hydroxide. Weight loss shall be not more than 0.144 percent when tested as specified in 4.6.3.

3.1.1.2.3 *Solubility.* No water marks shall remain on the inner surface of the evaporating dish and no pink color shall develop in the test solution when tested as specified in 4.6.4.

3.1.1.3 *Glow.* The porcelain body shall not crack and the glaze shall not crack when tested as specified in 4.6.5.

3.1.2 *Fused silica, opaque.* Type II evaporating dishes shall be a high quality silicon dioxide having a maximum coefficient of linear expansion of 8.0×10^{-7} cm/cm/1°C. from 0° to 1100°C.

3.1.2.1 *Composition.* Silicon dioxide content shall be a minimum of 99.7 percent when tested as specified in 4.6.6.

3.1.2.2 *Thermal shock resistance.* No cracking shall occur when dishes are tested as specified in 4.6.1.2.

3.1.2.3 *Devitrification.* No crystalization of glass shall occur when subjected to prolonged heat as specified in 4.6.7.

3.1.3 *Borosilicate glass.* Type III evaporating dishes shall be fabricated from glass conforming to DD-G-541, type I, class A.

3.1.3.1 *Thermal shock resistance.* No cracking shall occur when the dishes are tested as specified in 4.6.1.2.

3.2 Style and design.

3.2.1 *Shape.* Types I and II evaporating dishes shall have round or slightly flattened bottoms; type III evaporating dishes shall have flat bottoms. All types shall have sloping sides and a well formed pouring lip. The dishes shall pour evenly when tested as specified in 4.6.8.

TABLE I. *Dimensions and capacities of evaporating dishes*

Type	Capacity milliliter (ml.)	Dimensions		Type	Capacity (ml.)	Dimensions	
		Top inside diameter* millimeter (mm.)	Height* (mm.)			Top inside diameter* (mm.)	Height* (mm.)
I	35	60	24	I	525	162	51
I	70	75	30	I	1285	215	63
I	90	85	33	I	2100	265	80
I	120	90	37	II	200	110	45
I	150	100	42	III	90	80	45
I	250	115	45	III	195	105	55

* Tolerance ± 2 mm.

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3.2.2 *Dimensions and capacities.* Dimensions shall be as shown in table I. Minimum capacity shall be the value shown. Maximum capacity shall not exceed this value by more than 15 percent.

3.3 **Identification marking.** Each evaporating dish shall be permanently and legibly marked with the name or trademark of the manufacturer. Markings shall be legible after shock, and chemical tests have been performed.

3.4 **Workmanship.** Evaporating dishes shall be symmetrical in shape, uniform in size, and free from cracks, specks, crazing, and bubbles, and other irregularities which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1.1, 3.1.2, 3.1.3, and section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the evaporating dishes of the same type and same capacity offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions.

4.4 **Sampling.**

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For test.* Sample evaporating dishes shall be taken at random from each lot in accordance with the number of samples required for test as shown in table II.

TABLE II. *Number of random samples required.*

	Type			Reference paragraph
	I	II	III	
Thermal shock resistance	2	3	4	4.6.1
Glaze adhesion	4	—	—	4.6.2
Resistance to chemical attack	3	—	—	4.6.3
Solubility	3	—	—	4.6.4
Glow	2	—	—	4.6.5
Silicon dioxide	—	2	—	4.6.6
Devitrification	—	2	—	4.6.7
Pour test	2	2	2	4.6.8
Total number of samples required	16	9	6	

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4.5 **Examination.** Samples of evaporating dishes and level A and level B preparation for delivery thereof shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Evaporating dishes.*

Categories	*Defects*
Critical:	None defined
Major:	*AQL 4.0 percent defective*
101	Capacity, style, design, or dimensions incorrect
102	Damage (cracks, crazing, bubbles)
103	Contamination (grease, oil, or foreign matter)
104	Markings missing, incorrect, or illegible.

4.5.1.2—*Level A and level B preparation for delivery* (*section* 5).

Categories	*Defects*
Critical:	None defined
Major:	*AQL 1.5 percent defective*
101	Wrapping of unit, intermediate, or shipping container not as specified
102	Number of evaporating dishes per intermediate container not as specified
103	Unlike evaporating dishes (type or capacity) packaged per intermediate or shipping container
104	Closure of unit, intermediate, or shipping container not as specified
105	Unit, intermediate, or shipping container damaged
106	Cushioning material not as specified
107	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete

4.6 **Tests.** Distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.6.1 *Thermal shock resistance.*

4.6.1.1 *Type I.* Porcelain evaporating dishes shall be heated to 160° ± 2°C. in an oven for 30 minutes, removed, and immediately plunged into water at a temperature of 9° ± 1°C.

4.6.1.2 *Type II and type III.* Method 401, Fed. Test Method Std. No. 626 shall be used. Type II dishes shall be heated to 1000° ± 25°C.; type III dishes shall be heated to 180° ± 2°C.

4.6.2 Glaze adhesion (*type I, only*). Place two porcelain evaporating dishes so that glazed surfaces are in contact with each other, and heat to 1250°C. in an electric muffle-type furnace for 30 minutes. Allow to cool by radiation to room temperature.

4.6.3 *Resistance to chemical attack* (*type I, only*). Weighed evaporating dish shall be boiled in 10 percent solution of sodium hydroxide for 8 hours, rinsed, dried, and reweighed. Weight loss shall be calculated as follows:

$$L = \frac{100\ (B - A)}{B}$$

Where:

L = percent weight loss,

B = weight, before boiling, in grams, and

A = weight, after boiling, in grams.

4.6.4 *Solubility* (*type I, only*). Each evaporating dish, taken for test, shall be cleaned and rinsed three times with distilled water, using a new quantity of distilled water for each rinsing. The evaporating dishes shall be air-dried in an inverted position for 1 minute after each rinsing. After the final drying operation, each evaporating dish shall be half-filled with distilled water at 60°C. containing 0.2 percent phenolphthalein solu-

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tion (prepared from 1 gram phenolphthalein in 100 ml. alcohol) and autoclaved under 15 pounds steam pressure for 30 minutes. The distilled water used in this test shall have just been boiled for 10 minutes and allowed to cool to 60°C., prior to the addition of the phenolphthalein solution.

Note. Glassware used in this test shall not affect test solution.

4.6.5 *Glow (type I, only).* Porcelain-evaporating dishes shall be heated to bright red (approximately 1000°C) in an electric muffle-type furnace, then removed and allowed to cool by radiation to room temperature.

4.6.6 *Silicon dioxide content (type II only)* Applicable sections of ASTM No. C146-43 shall be used.

4.6.7 *Devitrification (type II only).* Test samples shall be heated to 1100°C in an electric muffle-type furnace for 8 hours and allowed to cool in place overnight

4.6.8 *Pour test.* Fill the dish to indicated capacity with water. Pour approximately one-fourth of the water from the dish at such a rate that the water flows in a continuous stream and falls free from the lip of the dish. If the stream of water clings to the outside of the dish and flows off the bottom, it shall be considered cause for rejection.

4.7 **Acceptance/rejection criteria.** If any evaporating dish fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1. *Level A.* Each evaporating dish shall be wrapped in 30-pound basis weight-kraft paper conforming to grade B of UU-P-268. The wrapping shall be secured with tape conforming to UU-T-111.

5.1.1.1 *Intermediate packaging.* Each dish, or the number of dishes specified (see 6.2) of the same type and capacity, shall be packaged in a fiber box conforming to type I, class 1, style optional, of PPP-B-636. The box shall be provided with suitable interior fittings or shall contain sufficient cushioning material conforming to type II, class B, of PPP-C-843 to prevent shifting of the contents. The box shall be closed in accordance with the appendix of the box specification.

5.1.2 *Level C.* Evaporating dishes shall be packaged in accordance with manufacturer's commercial practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Evaporating dishes shall be packed in a shipping container conforming to PPP-B-621, class 2; PPP-B-601, overseas type; or PPP-B-636, type I, class 2 or type II, class 2. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare consistant with the protection required, and contain identical quantities of like items. The gross weight of each pack shall not exceed 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547 which shall be sealed in accordance with the appendix thereto. PPP-B-636 boxes sealed at all joints and seams with tape conforming to PPP-T-60 shall not require case liners.

5.2.2 *Level B.* Evaporating dishes shall be packed in accordance with 5.2.1 except that shipping containers shall conform to PPP-B-636, type I or II, class 1, style RSC, minimum bursting strength 275 pounds per square inch; PPP-B-621, class 1; PPP-B-601, domestic type; PPP-B-591, domestic type;

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or PPP-B-576, domestic type. Case liners shall not be required.

5.2.3 *Level C.* Evaporating dishes shall be packed to insure carrier acceptance and safe delivery at first domestic destination for immediate use. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The evaporating dishes covered by this specification are intended for general laboratory use.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(b) Level of packaging and packing required.

(*c*) Type and capacity required.

(*d*) Number of dishes per intermediate and shipping container required.

6.3 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard and military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 **Transportation description.** Transportation description applicable to this item is:

Porcelainware, not otherwise indexed by name. (Specify valuation.)

Carload minimum weights 20,000 and 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Porcelainware, not otherwise indexed. (Specify valuation.) Motor volume minimum weight 24,000 pounds.

Glassware, chemical laboratory.
Carload minimum weight 18,000 pounds, subject to Rule 34, Uniform Freight Classification.

Glassware, laboratory.
Motor volume minimum weight 18,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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